

Kiste Phase 8.5.1 Spec

Data/Blob Storage Revision for Release Stabilization

One-sentence spec: Kiste Phase 8.5.1 revises Kiste data/blob handling so large datasets, model files, logs, media, and artifacts are represented by metadata, references, manifests, checksums, access rules, cost warnings, and export plans instead of being downloaded or committed by default.

Phase	Main focus
Phase 5	Python library + CLI stabilization: public API, installable package, CLI wrapper.
Phase 6	Many repo providers, Docker Hub/image inspection, and serverless/Kubernetes target planning.
Phase 7	Dataset intelligence: Kaggle, Roboflow, Hugging Face Datasets, DVC, local folders, and cloud data sources.
Phase 8	Self-hosting, modular bootstrap, cloud readiness, and Argo CD preparation.
Phase 8.5	Release stabilization: API freeze, CLI freeze, tests, docs, security, bootstrap validation.
Phase 8.5.1	Data/blob storage revision: metadata-first blobs, small sampling, manifests, access checks, and safe downloads.

1. Purpose

Phase 8.5.1 stays inside the stabilization track. It does not introduce a full data lake, a new storage platform, or automatic large downloads. Its job is to make Kiste release-safe when projects include big datasets, model weights, media files, logs, and binary artifacts.

The key change is that Kiste treats blob storage as external truth. Kiste records where large data lives, how to access it, how to verify it, how to sample it safely, and how to reproduce it later - without copying everything into Git or downloading it by default.

2. Core Principles

Principle	Meaning
Metadata first	List objects, sizes, timestamps, formats, versions, and policies before reading content.
References, not blobs	GitOps and .kiste state should store BlobRef manifests, not raw data archives or model weights.
Explicit download	Large blob downloads require an explicit command and confirmation.
Small sample allowed	Kiste may sample a small file or a bounded number of tiny files when needed for schema/format detection.
Cost-aware	Warn about egress, cross-region transfer, storage-class retrieval, and request costs.
Least privilege	Metadata read should be enough for normal scans. Delete/write permissions are not needed by default.
Reproducible	Manifests and checksums should make training, evaluation, and deployment reproducible.

3. What Counts as Blob Storage

- Object storage: S3, GCS, Azure Blob, MinIO, Cloudflare R2, Wasabi, and generic S3-compatible stores.
- Dataset platforms: Kaggle, Roboflow exports, Hugging Face Datasets, and DVC remotes.
- Model artifact stores: Hugging Face model files, safetensors, ONNX files, checkpoints, and weights.
- Local development data: local folders, sample datasets, test fixtures, and small CSV/image folders.
- Large binary artifacts: videos, images, logs, archives, generated model outputs, and benchmark artifacts.

4. BlobRef Model

Add a normalized **BlobRef** object. DatasetSpec explains what the dataset means; BlobRef explains where the physical data lives and how Kiste may access it.

```
{
  "name": "weld-seam-images",
  "kind": "dataset",
  "provider": "s3",
  "uri": "s3://kiste-datasets/weld-seams/v1/",
  "version": "v1",
  "content_type": "image-dataset",
  "format": "yolov8",
  "size": { "files": 12000, "bytes": 8420000000 },
  "checksum": { "type": "manifest-sha256", "value": "abc123..." },
  "access": {
    "token": "aws-data-dev",
    "permissions": ["blob:metadata:read", "blob:object:read"],
    "private": true
  },
  "policy": {
    "download_by_default": false,
    "max_scan_bytes": 10485760,
    "allow_sample": true,
    "max_sample_files": 20,
    "max_sample_file_bytes": 1048576
  }
}
```

5. Relationship Between kiste data and kiste blob

kiste data handles dataset meaning: task type, classes, splits, labels, license, lineage, and intended use. **kiste blob** handles physical storage: URI, files, size, checksums, access, sampling, and export/download plans.

```
datasets:
- name: weld-detection
  provider: roboflow
  task_type: object-detection
  blobs:
    - name: weld-detection-yolov8-export
      provider: roboflow
      uri: roboflow://workspace/project/1/yolov8
    - name: weld-detection-backup
      provider: s3
      uri: s3://kiste-datasets/weld-detection/v1/
```

6. New Command Group

Phase 8.5.1 adds a focused blob command group. It should be safe by default and compatible with release checks.

```
kiste blob add local ./datasets/weld-seams --name weld-seams
kiste blob add s3 s3://kiste-datasets/weld-seams/v1/ --name weld-seams-s3
kiste blob inspect weld-seams
kiste blob scan weld-seams
kiste blob manifest weld-seams
kiste blob check weld-seams
kiste blob sample weld-seams --limit 20
kiste blob export-plan weld-seams
kiste blob download weld-seams --confirm
kiste release check --include-blobs
```

7. Metadata-First Scan

A blob scan should first collect cheap metadata. It should avoid full object reads unless the user explicitly asks for deep verification or download.

- List object names or file paths.
- Read object sizes and modified timestamps.

- Detect extensions, folder structure, and split folders such as train/valid/test.
- Detect obvious metadata files such as data.yaml, labels.json, dataset card, README, manifest, or DVC files.
- Estimate total size, object count, and format hints.
- Generate a manifest checksum based on metadata, not full content, unless deep mode is requested.

```

Blob scan: weld-seams

[OK] Provider: s3
[OK] URI: s3://kiste-datasets/weld-seams/v1/
[OK] Objects found: 12,000
[OK] Total size: 8.42 GB
[OK] Format hints: jpg, png, txt, yaml
[OK] Split folders: train, valid, test
[WARN] Full checksum requires reading all objects
[WARN] Download disabled by default
[OK] Manifest generated: .kiste/blobs/weld-seams.manifest.json

```

8. Small Sample Policy

Kiste may sample a small file when metadata alone is not enough. This is useful for detecting schema, annotation format, image dimensions, CSV columns, JSONL shape, or whether labels match expected classes.

Sampling is allowed only with strict limits. It must not become a hidden bulk download.

Rule	Requirement
Bounded count	Default max_sample_files should be small, for example 20.
Bounded size	Default max_sample_file_bytes should be small, for example 1 MB per file.
Prefer metadata files	Read README, data.yaml, labels.json, schema.json, or manifest files before media blobs.
No raw sample persistence by default	Store sample-derived metadata, not the sampled file content, unless --save-sample is explicit.
PII caution	If text/tabular samples may contain personal data, record a warning and avoid printing raw rows by default.
Audit trail	Record which paths were sampled, their sizes, hashes, and reason for sampling.
User override	Allow --no-sample for strict metadata-only operation.

```

kiste blob sample weld-seams --limit 20 --max-file-bytes 1048576
kiste blob scan weld-seams --no-sample
kiste blob sample weld-seams --metadata-only

```

```

Sample output:
[OK] Sampled 12 metadata/small files
[OK] Detected YOLO label format
[OK] Detected image dimensions from 5 small images
[WARN] Raw samples were not saved
[OK] Sample audit written: .kiste/blobs/weld-seams.sample.json

```

9. Manifest Format

Kiste should generate a blob manifest that is small enough to commit, review, and compare. The manifest records enough information for reproducibility and access checks without containing the dataset itself.

```

{
  "blob": {
    "name": "weld-seams",
    "provider": "s3",
    "uri": "s3://kiste-datasets/weld-seams/v1/",
    "scanned_at": "2026-05-23T00:00:00Z"
  },
  "summary": {
    "objects": 12000,
    "bytes": 8420000000,
    "extensions": [".jpg", ".png", ".txt", ".yaml"]
  },
}

```

```

"objects_sample": [
  {
    "path": "train/images/0001.jpg",
    "size": 204812,
    "modified": "2026-05-01T12:00:00Z"
  }
],
"splits": { "train": 9000, "valid": 2000, "test": 1000 },
"checksum": { "type": "manifest-sha256", "value": "abc123..." }
}

```

10. Access and Permission Model

Blob storage uses the same capability-based token system as repo and data providers. Normal scans should require metadata read only. Object read is needed for sampling or downloading. Delete permission should almost never be required.

```
kiste token add aws --name aws-data-dev --scope blob:metadata:read,blob:object:read --expires 2026-08-01
```

- blob:metadata:read - list objects and metadata.
- blob:object:read - read small samples or download with confirmation.
- blob:object:write - write exports or generated artifacts, not needed by default.
- blob:delete - dangerous, should not be requested by default.
- blob:sign-url - generate temporary signed URLs when supported.
- blob:inventory:read - read provider inventory manifests when available.

11. Cost and Safety Checks

Blob operations can create cost through egress, API requests, storage class retrieval, cross-region transfer, and GPU training data movement. Kiste should warn before any operation that may move many bytes.

```

kiste blob export-plan weld-seams

Export plan: weld-seams
[OK] Source: s3://kiste-datasets/weld-seams/v1/
[OK] Objects: 12,000
[OK] Estimated size: 8.42 GB
[WARN] Cross-region download may create egress cost
[WARN] Full download requires explicit --confirm
[OK] Recommended: metadata scan + small sample first

```

12. Workspace and GitOps Integration

Phase 8.5.1 adds a blob graph layer to the existing repo, image, target, and dataset graph. GitOps outputs should store BlobRef YAML, not raw data.

```

training-repo -> dataset:weld-detection
dataset:weld-detection -> blob:s3://kiste-datasets/weld-seams/v1/
model-repo -> blob:hf://owner/model/model.safetensors
api -> image:docker.io/owner/api:latest

kiste workspace graph --include-data --include-blobs
kiste workspace check --include-data --include-blobs

apiVersion: kiste.dev/v1alpha1
kind: BlobRef
metadata:
  name: weld-seams-v1
spec:
  provider: s3
  uri: s3://kiste-datasets/weld-seams/v1/
  version: v1
  checksum:
    type: manifest-sha256
    value: abc123
  access:

```

```
secretRef: aws-data-dev
policy:
  downloadByDefault: false
  maxSampleFiles: 20
  maxSampleFileBytes: 1048576
```

13. Release Checks

Phase 8.5.1 must feed into release stabilization. Kiste should reject unsafe blob behavior before release.

```
kiste release check --include-blobs

Blob storage release check:
[OK] BlobRef model exists
[OK] Blob manifest generation works
[OK] Small sample limits are enforced
[OK] Blob download requires --confirm
[OK] GitOps output stores references only
[WARN] S3 provider uses metadata-only scan
[FAIL] Example dataset has no checksum manifest
```

14. What Phase 8.5.1 Should Not Do

- No full data lake platform.
- No automatic huge dataset download.
- No automatic bucket/container creation.
- No automatic object deletion.
- No full PII detector or legal/license engine.
- No automatic model retraining.
- No replacement for DVC, S3, GCS, Azure Blob, Hugging Face, Kaggle, or Roboflow.

15. Acceptance Criteria

Phase 8.5.1 is complete when these commands work safely from a clean workspace:

```
kiste blob add local ./datasets/weld-seams --name weld-seams
kiste blob add s3 s3://kiste-datasets/weld-seams/v1/ --name weld-seams-s3
kiste blob inspect weld-seams
kiste blob scan weld-seams
kiste blob manifest weld-seams
kiste blob check weld-seams
kiste blob sample weld-seams --limit 20
kiste blob export-plan weld-seams
kiste workspace graph --include-data --include-blobs
kiste workspace check --include-data --include-blobs
kiste release check --include-blobs
```

Required output	Meaning
BlobRef model	Normalized reference for large data and artifacts.
Blob provider abstraction	Local, object store, dataset platform, and DVC-style providers.
Metadata-only scan	Safe scan that avoids full content reads.
Small sample support	Bounded sampling for schema/format detection when needed.
Blob manifest file	Portable manifest with summary, sample paths, and checksums.
Access check	Capability check for metadata read, object read, and unsafe permissions.
Export/download plan	Cost-aware and confirmation-gated plan.
Workspace blob graph	Connections between repos, datasets, models, images, and blobs.
GitOps-safe BlobRef YAML	Reference manifests only, no raw blob content.

Release check	Detect unsafe downloads, large files in Git, missing checksums, and weak policies.
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16. Final Rule

Kiste should know where large data lives, how to access it, how risky it is, how to sample it safely, and how to reproduce it - without downloading or committing it by default.

This document is a Phase 8.5.1 stabilization addendum. It revises blob/data handling before release so Kiste remains safe, modular, reproducible, and cost-aware.